"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

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q1

a) $p' \vee q' \vee p \equiv (p' \vee p) \vee q' \equiv \perp$

b) $p' \vee (p \vee q) \equiv (p' \vee p) \vee q \equiv \perp$

c) $p \vee (p' \vee q) \equiv (p \vee p') \vee q \equiv \perp$

d) $(p' \vee q') \vee (p' \vee q) \equiv (q' \vee q) \vee (p' \vee p') \equiv \perp \vee (p' \vee p') \equiv \perp$

p	q	p∧q	p∧q→p
⊥	⊥	⊥	⊥
⊥	0	0	⊥
0	⊥	0	⊥
0	0	0	⊥

↳ tautology

p	q	p∨q	p→p∨q
⊥	⊥	⊥	⊥
⊥	0	⊥	⊥
0	⊥	⊥	⊥
0	0	0	⊥

↳ tautology

p	q	p'	p→q	p'→(p→q)
⊥	⊥	0	⊥	⊥
⊥	0	0	0	⊥
0	⊥	⊥	⊥	⊥
0	0	⊥	⊥	⊥

↳ tautology

p	q	p∧q	p→q	(p∧q)→(p→q)
⊥	⊥	⊥	⊥	⊥
⊥	0	0	0	⊥
0	⊥	0	⊥	⊥
0	0	0	⊥	⊥

↳ tautology

q2

$f+g = f(x)+g(x) = x^2+1+x+2 = x^2+x+3$

$f_g = f(x).g(x) = (x^2+1).(x+2) = x^3+2x^2+x+2$

q3

Boy Boy Boy Boy Boy Boy Boy Boy Boy Boy

There are eleven place for women stand so first,

10 boy we use $P(10,10) = \frac{10!}{(10-10)!} = 10!$

Also we are using permutation because order is important

then

we can seperate women $P(11,6) = \frac{11!}{(11-6)!} = \frac{11!}{5!}$ (11 place, 6 woman)

Finally the result $= \frac{10! \cdot 11!}{5!} = 1\,207\,084\,032\,000$

using calculator

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9.4

a) n cars are made in the nth month. Let a_n as number of cars made in the first n months.

$$a_n = a_{n-1} + n$$, so 0 cars are made in the 0th month. then $a_0 = 0$

b) $a_n = a_{n-1} + n$

$= (a_{n-2} + n-1) + n = a$

$= (a_{n-3} + n-2) + (n + (n-1))$

$\vdots$

$$a_0 + \sum_{i=1}^{n} i$$

$$= 0 + \frac{n(n+1)}{2}$$

$$= a_n = \frac{n(n+1)}{2}$$

c) As we know $a_n = \frac{n(n+1)}{2}$, $n = 12$.

$$a_{12} = \frac{12 \cdot 13}{2} = 78 \;//$$